

WEEK 2

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using `fork()` system call.
- b) Read the user input as shell command.
- c) Use a system call `exec()` to execute the shell command.
- d) Use a `wait()` system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
imvarls@LAPTOP-VQ4D70L7:~$ nano file.c|
```

```
imvarls@LAPTOP-VQ4D70L7:~$ nano file.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc file.c -o file
imvarls@LAPTOP-VQ4D70L7:~$ ./file|
```

```
imvarls@LAPTOP-VQ4D70L7:~$ nano file.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc file.c -o file
imvarls@LAPTOP-VQ4D70L7:~$ ./file
=== Simple Linux Command Executor ===
Enter a Linux command (e.g., 'ls -l', 'date', 'pwd'): |
```

```
imvarls@LAPTOP-VQ4D70L7:~$ nano file.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc file.c -o file
imvarls@LAPTOP-VQ4D70L7:~$ ./file
=== Simple Linux Command Executor ===
Enter a Linux command (e.g., 'ls -l', 'date', 'pwd'): pwd

[Parent Process] PID: 1022
[Parent Process] Waiting for child (PID: 1025) to finish...
[Child Process] PID: 1025 | Parent PID: 1022
[Child Process] Executing command: 'pwd'...

/home/imvarls

[Parent Process] Child completed. Parent exiting now.
imvarls@LAPTOP-VQ4D70L7:~$ |
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
imvarls@LAPTOP-VQ4D70L7:~$ uname
Linux
imvarls@LAPTOP-VQ4D70L7:~$ |
```

```

imvarls@LAPTOP-VQ4D70L7:~$ uname
Linux
imvarls@LAPTOP-VQ4D70L7:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) 5 120U
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               3
BogoMIPS:               4992.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdp1gb rdtscp lm constant
                        _tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                        opcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                        d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                        vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_lid arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
  L1d:                   288 KiB (6 instances)
  L1i:                   192 KiB (6 instances)
  L2:                     7.5 MiB (6 instances)
  L3:                     12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-11
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:           Not affected
L1tf:                    Not affected
Mds:                     Not affected
Meltdown:                Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Vulnerable: No microcode
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                   Not affected
Tsa:                     Not affected
Tsx async abort:         Not affected
Vmscape:                 Not affected
imvarls@LAPTOP-VQ4D70L7:~$

```

```

imvarls@LAPTOP-VQ4D70L7:~$ ps
  PID TTY          TIME CMD
  778 pts/0        00:00:00 bash
 1114 pts/0        00:00:00 ps
imvarls@LAPTOP-VQ4D70L7:~$ |

```

imvarls@LAPTOP-VQ4D70L7:~\$ top

top - 15:43:22 up 50 min, 1 user, load average: 0.08, 0.02, 0.01

Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 7782.6 total, 7094.6 free, 578.7 used, 260.4 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7203.9 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24552	15508	11592	S	0.0	0.2	0:00.78	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
7	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50396	16840	15568	S	0.0	0.2	0:00.29	systemd-journal
80	systemd+	20	0	22540	14636	12016	S	0.0	0.2	0:00.11	systemd-resolve
90	root	20	0	35364	12280	9184	S	0.0	0.2	0:00.76	systemd-udev
108	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.03	chronyd-starter
113	root	20	0	4472	3112	2868	S	0.0	0.0	0:00.05	cron
117	message+	20	0	8792	5456	4772	S	0.0	0.1	0:00.05	dbus-daemon
154	root	20	0	44092	29732	14592	S	0.0	0.4	0:00.21	networkd-dispat
158	root	20	0	18432	9444	8200	S	0.0	0.1	0:00.12	systemd-logind
185	syslog	20	0	220548	5968	4660	S	0.0	0.1	0:00.06	rsyslogd
209	_chrony	20	0	24440	11956	7184	S	0.0	0.2	0:00.16	chronyd
213	root	20	0	5492	2756	2572	S	0.0	0.0	0:00.01	agetty
226	_chrony	20	0	12092	2424	1784	S	0.0	0.0	0:00.00	chronyd
270	root	20	0	123456	32576	17964	S	0.0	0.4	0:00.15	unattended-upgr
322	root	20	0	8648	5540	4696	S	0.0	0.1	0:00.01	login
364	imvarls	20	0	22508	13432	10984	S	0.0	0.2	0:00.11	systemd
368	imvarls	20	0	22908	4080	2080	S	0.0	0.1	0:00.00	(sd-pam)
395	imvarls	20	0	6136	5412	3864	S	0.0	0.1	0:00.02	bash
774	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
775	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.09	Relay(778)
778	imvarls	20	0	6268	5544	3872	S	0.0	0.1	0:00.11	bash
1126	imvarls	20	0	8272	6068	3912	R	0.0	0.1	0:00.00	top